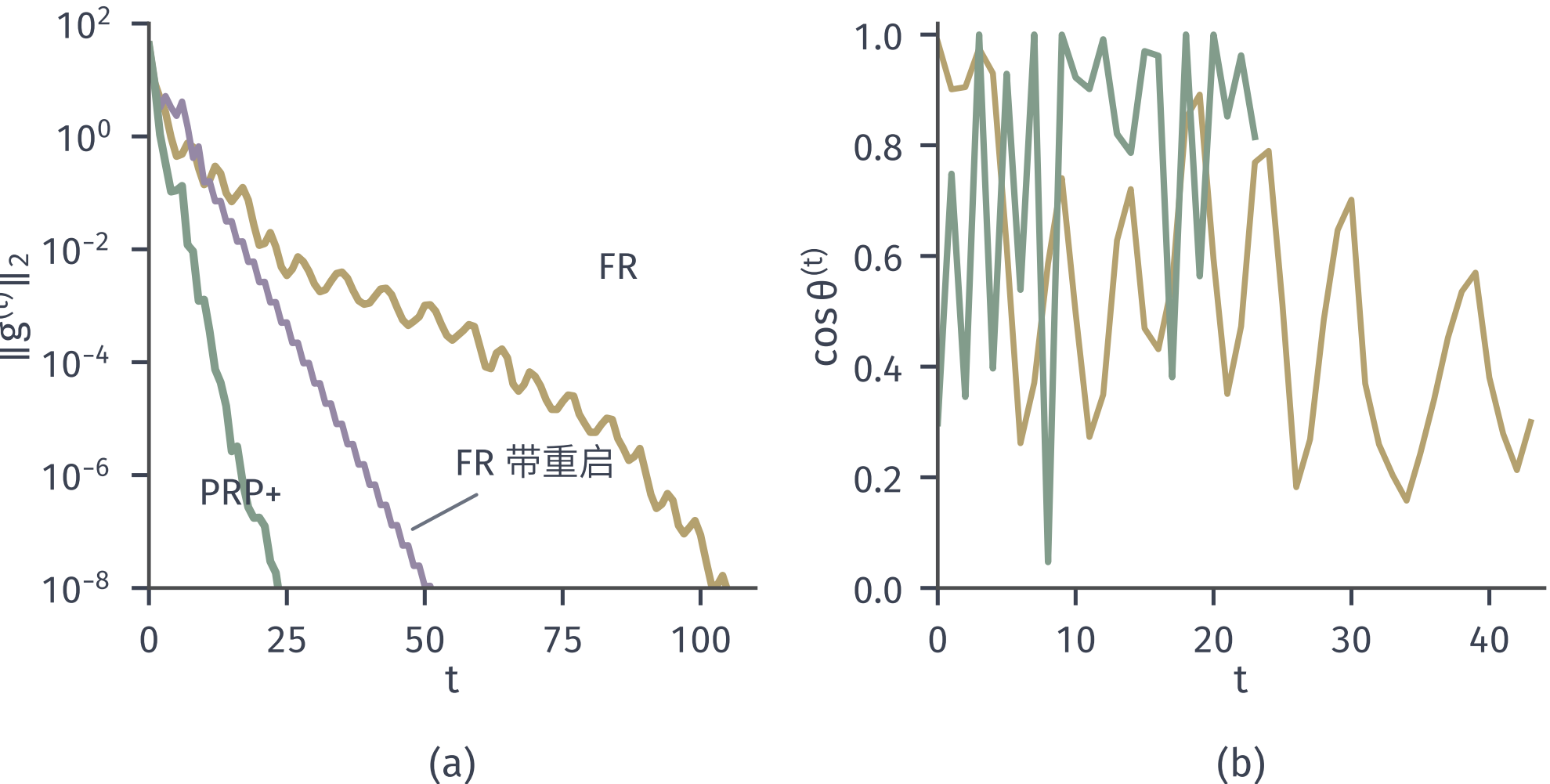
10^2
10^0
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$\|g^{(t)}\|_2$
FR
FR 带重启
PRP+
0
25
50
75
100
t
(a)
1.0
0.8
0.6
0.4
0.2
0.0
$\cos\theta^{(t)}$
0
10
20
30
40
t
(b)